

Voronoi stability from quantitative flat-direction transversality

Partial solution packet for arXiv:1212.1094

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Candidate partial result. Finite-face decomposition can be removed from the source’s stability theorem if cross-site directions stay a positive distance from the directions of flat segments in the unit sphere. Closedness of the flat-direction set makes the source’s original qualitative general-position condition sufficient and covers the cylinder norm in Example 3.5.

1 Source direction and notation

The source is Daniel Reem, *The geometric stability of Voronoi diagrams in normed spaces which are not uniformly convex*, arXiv:1212.1094v2 [1]. Section 8, PDF page 14, asks in particular to

“weaken the finite face decomposition assumption imposed on the structure of the unit sphere and to see whether a stability result can be formulated for the normed spaces induced by the unit spheres mentioned in Examples 3.4–3.5.”

Let $E = (\mathbb{R}^m, \|\cdot\|)$, let S be its unit sphere, and let $X \subset E$ be compact and convex. For nonempty sets $P, A \subset X$ with $d(P, A) > 0$, define the symmetric set of normalized cross-directions

$$\widehat{P, A} = \left\{ \pm \frac{a - p}{\|a - p\|} : p \in P, a \in A \right\}.$$

The flat-direction set of the norm is

$$\widehat{S} = \left\{ \frac{s' - s}{\|s' - s\|} : s \neq s', [s, s'] \subset S \right\}.$$

For sites $(P_k)_{k \in K}$, put $A_k = \bigcup_{j \neq k} P_j$ and

$$R_k = \text{dom}(P_k, A_k) := \{x \in X : d(x, P_k) \leq d(x, A_k)\}.$$

The Hausdorff distance is denoted by D .

2 Proof intuition

The source does not use individual faces throughout its long stability proof. It uses their finiteness once, to turn the qualitative statement “no site direction is flat” into a positive angular separation $2r$. Everything after that point is phrased only in terms of r . If the positive separation is assumed directly, small Hausdorff perturbations move normalized cross-site directions by at most $4\Delta/\eta$, so half the gap survives. The source’s uniform-margin lemma then applies simultaneously to the original and perturbed sites, and its abstract stability proposition finishes the proof.

3 Stability without finite faces

Theorem 1 (quantitative transversality). *Let $X \subset (\mathbb{R}^m, \|\cdot\|)$ be compact and convex, and let $(P_k)_{k \in K}$ be finitely many nonempty compact sites. Suppose*

$$\eta := \min_{j \neq k} d(P_j, P_k) > 0 \quad \text{and} \quad \rho := \min_{k \in K} \text{dist}(\widehat{P_k}, \widehat{A_k}, \widehat{S}) > 0.$$

Then for every $\epsilon > 0$ there is $\Delta > 0$ such that, whenever nonempty compact sites $(P'_k)_{k \in K}$ satisfy $D(P_k, P'_k) < \Delta$ for every k , their Voronoi cells satisfy

$$D(R_k, R'_k) < \epsilon \quad (k \in K).$$

The corresponding bisectors $B_k = \{x : d(x, P_k) = d(x, A_k)\}$ satisfy $D(B_k, B'_k) \leq \epsilon$ after decreasing Δ if necessary. No finite-face decomposition of S is assumed.

Proof. We isolate the exact substitution in the source proof. It is enough to treat $0 < \epsilon < \eta/6$; a smaller tolerance proves any larger one. Put $r = \rho/2$ and $M = \text{diam } X$.

The source's Lemma 9.15 states that, for fixed $\epsilon, r > 0$, there is a single $\lambda \in (0, \epsilon)$ giving its radial margin estimate for every compact pair (P, A) with

$$d(P, A) > 2\epsilon, \quad \text{dist}(\widehat{P}, \widehat{A}, \widehat{S}) \geq r.$$

This lemma is proved for an arbitrary finite-dimensional norm and does not use finite-face decomposition.

Choose $\Delta > 0$ so small that

$$\Delta < \min \left\{ \epsilon, \frac{\lambda}{8(1 + M/\epsilon)}, \frac{r\eta}{8} \right\}. \quad (1)$$

For $A'_k = \bigcup_{j \neq k} P'_j$, the elementary union estimate gives $D(A_k, A'_k) < \Delta$, and hence

$$d(P'_k, A'_k) \geq d(P_k, A_k) - 2\Delta > \eta - 2\epsilon > 2\epsilon.$$

The direction estimate in the source's Lemma 9.5 gives

$$D(\widehat{P_k}, \widehat{A_k}, \widehat{P'_k}, \widehat{A'_k}) \leq \frac{4\Delta}{\eta} < \frac{r}{2}.$$

Since the original gap is at least $\rho = 2r$, the perturbed gap is larger than $3r/2$, and in particular at least r . Thus Lemma 9.15 supplies the same radial margin λ for all original and perturbed pairs.

The boundedness requirement in the source's Proposition 9.2 holds with $M = M' = \text{diam } X$. Its remaining smallness condition is precisely ensured by the middle term of (1). Proposition 9.2 therefore gives $D(R_k, R'_k) < \epsilon$ for every k .

For bisectors, the positive gaps imply the source's no-parallel-segment condition for both the original and perturbed sites. Its Theorem 7.1 identifies each bisector with the cell boundary, and the proof of Corollary 5.2 converts the cell estimates just obtained into $D(B_k, B'_k) \leq \epsilon$. These arguments also do not use finite faces. $\square$

4 Qualitative criterion and the cylinder

Corollary 2 (closed flat-direction set). *Suppose $\widehat{S}$ is closed. Under the other hypotheses of Theorem 1, assume only the source's qualitative general-position condition: no segment joining different sites is parallel to a nondegenerate segment contained in S . Then the conclusions of Theorem 1 hold.*

More generally, closedness of $\widehat{S}$ can be replaced by

$$\widehat{P_k}, \widehat{A_k} \cap \widehat{\widehat{S}} = \emptyset \quad (k \in K).$$

Proof. Because the sites are compact and separated, the map $(p, a) \mapsto (a - p)/\|a - p\|$ is continuous on each compact product $P_k \times A_k$. Hence $\widehat{P_k, A_k}$ is compact. The qualitative condition says that it is disjoint from $\widehat{S}$. If $\widehat{S}$ is closed, compactness of the unit sphere gives positive distance. The same argument applies directly with $\widehat{\widehat{S}}$ in the more general formulation. Theorem 1 now applies. $\square$

Corollary 3 (cylinder norm). *On $E = \mathbb{R}^2 \times \mathbb{R}$ equip (u, z) with*

$$\|(u, z)\| = \max\{\|u\|_2, |z|\}.$$

For compact separated sites in a compact convex world, Voronoi cells and bisectors are stable whenever no cross-site segment is horizontal or vertical.

Proof. The unit ball is $B_2 \times [-1, 1]$. Any nondegenerate segment in its boundary lies in a non-singleton exposed face: take a supporting functional at the midpoint, and linearity forces both endpoints into its maximizing face.

A functional on this product has the form (v, c) . If $v \neq 0$ and $c \neq 0$, its maximizing face is a singleton, because the Euclidean disk is strictly convex and the interval maximum is unique. If $c = 0$, the non-singleton face is a vertical generator $\{v/\|v\|_2\} \times [-1, 1]$; if $v = 0$, it is a horizontal top or bottom disk $B_2 \times \{\pm 1\}$. Conversely these faces contain all vertical and horizontal directions. Therefore

$$\widehat{S} = \{(u, 0) : \|u\|_2 = 1\} \cup \{(0, 0, 1), (0, 0, -1)\},$$

a closed set. The stated hypothesis is exactly the source’s qualitative general-position condition, so Corollary 2 applies. $\square$

5 Scope, verification, and novelty

The theorem gives a stability formulation for arbitrary finite-dimensional norms, including the source’s infinitely faceted examples, provided site directions avoid the *closure* of the flat directions. It does not prove that pointwise avoidance of a nonclosed $\widehat{S}$ is sufficient; that is the remaining obstruction for the weakest possible generalization.

A source-level search confirms that finite-face decomposition enters the proof layer only in Lemmas 9.10 and 9.12, whose downstream output is the gap used at the first line of the proof of Theorem 5.1. The constants above were checked against that proof and Lemmas 9.5 and 9.15. The packet uses no computational evidence. Cheap run indexes and the parsed arXiv corpus found no later exact theorem under this replacement condition. Expert review is recommended, with attention to the dependency audit, preservation of the direction gap, and the cylinder face classification. The result is submitted as a substantial partial answer with moderate novelty confidence.

References

- [1] Daniel Reem, *The geometric stability of Voronoi diagrams in normed spaces which are not uniformly convex*, arXiv:1212.1094v2, 2013.

The stability property established in Theorem 5.1 is quite general in the sense that the norm is not restricted to be uniformly convex and the sites can have a pretty general form (they only need to be compact). However, in some aspects this result is quite restricted comparing to the one presented in [72] since there infinitely many sites were allowed, the sites and the world X were not assumed to be compact or bounded (but a certain boundedness condition was assumed), and the normed space could be even infinite dimensional. Moreover, explicit (dimension free) estimates were given there in contrast with the non explicit ones given here. In particular, it is not clear whether Δ given in Theorem 5.1 does not depend on the dimension. However, we feel that by strengthening our approach it is possible to achieve the above properties in more general normed spaces, but for this one has to find ways to obtain explicit estimates not based on compactness or finiteness arguments. In addition, we also feel that the “general position” assumption on the sites can be weakened, with some caution (see e.g., Section 4 for a related discussion). It will be interesting to establish results of this spirit. In particular, it will be interesting to weaken the finite face decomposition assumption imposed on the structure of the unit sphere and to see whether a stability result can be formulated for the normed spaces induced by the unit spheres mentioned in Examples 3.4-3.5.

It is also interesting to generalize the results to other settings, such as manifolds, weighted distances, and convex distance functions (Minkowski functionals). Finding theoretical and real-world applications of the result (in addition to the possible ones described in Section 1) may be of interest too; a promising place where such a result

Figure 1: The finite-face weakening direction in Section 8, page 14 of arXiv:1212.1094v2.